



Semester 1 Examinations 2013 / 2014

Exam Code(s) 1SD1/1SD3/1MF1/1IT1
Exam(s) Higher Diploma in Applied Science (Software Design & Development), Higher Diploma in Applied Science (Software Design & Development) Industry Stream, M.Sc. in Software Design & Development, Master of Information Technology

Module Code(s) CT874
Module(s) Programming I

Paper No. I

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Instructions: Candidates are required to answer:

Section A (20 Marks) Answer all 10 parts of section A. This section consists of 10 multiple-choice questions, all of which should be attempted. **Answers are to be written on the MCQ sheet provided, NOT on the Examination paper.**

Section B (80 Marks) Answer Any **Two** Questions.

Duration 2 hours
No. of Pages 8
Department(s) Information Technology

Requirements MCQ

Section A (20 Marks)
Attempt all 10 questions
Mark your answers on the MCQ sheet provided

Q1.

- 1) If Motorcycle and Truck are subclasses of Vehicle. Which of these declarations are invalid?

I Truck t = new Vehicle();
II Vehicle v = new Truck();
III Motorcycle m1 = new Vehicle();
IV Motorcycle m2 = new Truck();

- a) I, III, and IV
- b) II and III
- c) II and IV
- d) I, II, III, and IV

- 2) What is wrong with the following code?

```
class super {  
    public int a;  
    protected int b;  
    private int c;  
}
```

- a) the class lacks a constructor
- b) private member variables are disallowed
- c) there is no such a thing as a protected access modifier
- d) super is a reserved word and should not be used.

- 3) What is the output of the following code?

```
int[] a = new int[20];  
a[0]=12;  
  
System.out.println(a.length);
```

- a) 0
- b) 1
- c) 12
- d) 20

4) What does the following program do?

```
double x = 0, unknown;

for (int j = 0; j < someArray.length; j++){
    x+=someArray[j];
}

unknown = x /someArray.length;
```

- a) Computes the sum of the doubles in the array, someArray.
- b) Deletes all the doubles in the array, someArray.
- c) Computes the average of all the doubles in the array, someArray.
- d) Computes the product of all the doubles in the array, someArray.

5) What happens when the following code is run?

```
String a = "cigarette";

for (int i = 0; i < 10; i++) {
    System.out.print(a.charAt(i));
}
```

- a) The output will be "c i g a r e t t e"
- b) The output will be c i g a r e t t e
- c) The output will be c i g a r e t t
- d) An exception will be thrown

6) What does the call, "super()" do?

- a) It will do nothing.
- b) Call the subclass's constructor
- c) Call the superclass's constructor.
- d) This is an invalid call.

7) We wish to make the following method handle exceptions:

```
public int getAge(){
    return this.Age();
}
```

How would we accomplish this?

- a)

```
throw{public int getAge(){
    return this.Age();
} catch (Exception e){ }}
```
- b)

```
try{public int getAge(){
    return this.Age();
} catch (Exception e){ }}
```
- c)

```
public int getAge(){
    throw{ return this.Age();
} catch (Exception e){ }}
```
- d)

```
public int getAge(){
    try{ return this.Age();
} catch (Exception e){ }}
```

8) Assume we wanted to make a constructor method in the class, "Nut".

```
class Nut{ String typeOfNut;

    public void Nut(String typeOfNut){
        -----;
    }
}
```

How would you set the class variable to the argument passed into Nut? Fill in the blank.

- a) `typeOfNut = typeOfNut;`
- b) `this = typeOfNut;`
- c) `this.typeOfNut = typeOfNut;`
- d) You must change the name of the parameter for this constructor to work properly.

9) What will happen when the following code is executed?

```
int i = 1;
while (i != 10)
    i = i*3;
```

- a) the program will produce an overflow error.
- b) the program will loop infinitely.
- c) a type mismatch error will occur.
- d) the program will complete normally

10) Consider the following program:

```
for (int i = 1; i <= 10; i++)
    for (int j = 1; j <= 10; j++)
        System.out.println("1");
```

How many lines of output will be printed?

- a) 1
- b) 10
- c) 20
- d) 100

Section B (80 Marks)
Answer Any TWO Questions

Q 2.

- a) In relation to object-oriented programming with Java, detail your understanding, using examples/analogies where appropriate, of each of the following:

- i. Abstract class
- ii. Interface
- iii. Class variable
- iv. Composition

[10]

- b) Write a Java application, which allows the user to input a line of text, tokenizes the string using an appropriate String method and outputs the tokens in reverse order. Use space characters as delimiters.

[15]

- c) Create a Java application that takes integer values from a user and determines whether each of the numbers inputted is odd or even. The application should include an ***isEven*** method, which takes an integer argument and returns true if the integer is even and false otherwise.

[15]

Q 3.

- a) Explain the terms **overloading** and **overriding** as used in object-oriented programming. You should provide snippets of Java code to illustrate your answer.

[10]

- b) Using the following interface, you are required to develop a Java application which will allow you to collect and store contact details.

```
public interface InterfaceContacts {
    String getLastName( );
    String getFirstName( );
    String getPhoneNumber( );
    String getEmailAddress( );
    void setLastName( String name );
    void setFirstName( String name );
    void setPhoneNumber( String phone );
    void setEmailAddress( String email );
}
```

You are required to:

- i. Create a **Contacts** class using the above interface as a specification for the desired methods.

[10]

- ii. Write a **test class** for the *Contacts* class called **ContactTest** and store the instances of *Contacts* created into a **Linked List**. The **ContactTest** class should implement the **addition** and **removal** of contacts to the Linked List and **display** its contents.

[20]

Q 4.

ACME Films require a Java application to keep track of BlueRay rentals. You are required to:

- a) Create a **Movie** class which has the following data members;

```
idNumber (static int)
title (String)
classification (String)
```

These data members are tracked using appropriate **setter** and **getter** methods. The classifications are in line with the Irish Film Classification Office and are as follows: G, PG, 12A, 15A, 16 and 18.

You are also required to appropriately plan for a **calculateLateFee** method, which **must** be overridden by classes derived from Movie. The method signature is as follows:

```
double calculateLateFee(int numDaysLate);
```

Each film is to have an **ID number**, a **title** and a **classification**.

[15]

- b) You are also required to create a **Drama** class, an **Action** class and a **Comedy** class that are *derived* from Movie. In each of these classes, create an overwritten method named **calculateLateFee**, which takes as input the number of days a film is late and returns the late fee for that film. The default late fee per day is €6, action films have a late fee of €5 per day, comedies are €4 per day and dramas are €3 per day.

[15]

- c) You are required to write a **test class** called **MovieTest** for your classes and the **MovieTest** class should store instances of Drama, Action and Comedy classes into an **Array List**.

[10]